Homework 00.11.26 – Exercises 31–37

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Exercise 31 (Decomposition of Bregman divergence)

Let $f = \frac{1}{n}\sum_{i=1}^n f_i$. Prove that

$$D_f(x,y) = \frac{1}{n}\sum_{i=1}^n D_{f_i}(x,y).$$

Proof

$$\begin{aligned} D_f(x,y) &= f(x) - f(y) - \langle \nabla f(y), x-y\rangle = \frac{1}{n}\sum_{i=1}^n f_i(x) - \frac{1}{n}\sum_{i=1}^n f_i(y) - \left\langle \frac{1}{n}\sum_{i=1}^n \nabla f_i(y), x-y\right\rangle \\ &= \frac{1}{n}\sum_{i=1}^n \underbrace{\left(f_i(x) - f_i(y) - \langle \nabla f_i(y), x-y\rangle\right)}_{D_{f_i}(x,y)} = \frac{1}{n}\sum_{i=1}^n D_{f_i}(x,y). \end{aligned}$$

Exercise 32 (Theorem 38)

Assume that

1. f is μ-convex.
2. Each f_i is convex and L_i-smooth.

Then SGD-NS with stepsize $0 < \gamma \le \frac{1}{2A''}$, where $A'' = \max_i \frac{L_i}{np_i}$, satisfies

$$\mathbb{E}\left[\left\|x^k - x^\star\right\|^2\right] \le (1-\gamma\mu)^k \left\|x^0 - x^\star\right\|^2 + \frac{2\gamma\sigma_\star^2}{\mu},$$

where

$$\sigma_\star^2 = \operatorname{Var}\left[g(x^\star)\right] = \mathbb{E}\left[\|g(x^\star)\|^2\right] - \left\|\mathbb{E}\left[g(x^\star)\right]\right\|^2 = \left(\frac{1}{n^2}\sum_{i=1}^n \frac{\|\nabla f_i(x^\star)\|^2}{p_i}\right) - \|\nabla f(x^\star)\|^2.$$

Proof

From Lemma 37, using the convexity and L_i-smoothness of f_i, we have that

$$\mathbb{E}\left[\|g(x) - g(y)\|^2\right] \le 2A''\, D_f(x,y), \qquad \text{where } A'' = \max_i \frac{L_i}{np_i}.$$

Moreover,

$$G(x,y) \le 4A''\, D_f(x,y) + 2\operatorname{Var}\left[g(y)\right] \qquad \forall x, y \in \mathbb{R}^d.$$

Furthermore, we use Theorem 33, which says that if f is μ-convex, g^k is unbiased and the AC assumption is satisfied, then choosing some $0 < \gamma \leq \frac{1}{A}$ we will have

$$\mathbb{E}\left[\left\|x^k - x^\star\right\|^2\right] \leq (1-\gamma\mu)^k \left\|x^0 - x^\star\right\|^2 + \frac{\gamma C}{\mu}.$$

In other words, Lemma 37 guarantees that the AC assumption is satisfied with

$$A = 2A'' = 2\max_i \frac{L_i}{np_i} \qquad \text{and} \qquad C = 2\,\mathrm{Var}\left[g(x^\star)\right] = 2\sigma_\star^2,$$

which can be used in Theorem 33, and after that γ can be chosen as $0 < \gamma \leq \frac{1}{A} = \frac{1}{2A''}$ to get

$$\mathbb{E}\left[\left\|x^k - x^\star\right\|^2\right] \leq (1-\gamma\mu)^k \left\|x^0 - x^\star\right\|^2 + \frac{2\gamma\sigma_\star^2}{\mu}.$$

Exercise 33

Prove that

$$\sigma_\star^2 = \mathbb{E}\left[\left\|g(x^\star) - \nabla f(x^\star)\right\|^2\right] = 0 \quad \text{iff} \quad \nabla f_i(x^\star) = np_i \nabla f(x^\star) \quad \forall i = 1, \ldots, n.$$

Proof

1)

$$\begin{aligned}\left(\frac{1}{n^2}\sum_{i=1}^n \frac{\left\|\nabla f_i(x^\star)\right\|^2}{p_i}\right) - \left\|\nabla f(x^\star)\right\|^2 &= \left(\frac{1}{n^2}\sum_{i=1}^n \frac{\left\|np_i\nabla f(x^\star)\right\|^2}{p_i}\right) - \left\|\nabla f(x^\star)\right\|^2 \\ &= \underbrace{\left(\sum_{i=1}^n p_i\right)}_{1} \left\|\nabla f(x^\star)\right\|^2 - \left\|\nabla f(x^\star)\right\|^2 = 0 \;\Rightarrow\; \sigma_\star^2 = 0.\end{aligned}$$

2) Let $\sigma_\star^2 = 0$, and let's assume that there are $p_i' > 0$ such that $\sum_i p_i' = 1$ and $\nabla f_i(x^\star) = np_i'\nabla f(x^\star)$. Then

$$\sigma_\star^2 = \left\|\nabla f(x^\star)\right\|^2 \left(\frac{1}{n^2}\sum_{i=1}^n \frac{(p_i'n)^2}{p_i} - 1\right) = 0 \iff \sum_{i=1}^n \frac{p_i'^2}{p_i} = 1.$$

One such solution for p_i' is $p_i' = p_i$. It also turns out that it is the only solution: just use Exercise 38 with $a_i := p_i'^2$ and $p_i := p_i$.

Exercise 34

Prove part (iii) of Lemma 44.

Lemma 44. Let $f_1, \ldots, f_n : \mathbb{R}^d \to \mathbb{R}$ be differentiable and $f(x) := \frac{1}{n}\sum_{i=1}^n f_i(x)$.

(i) If f_i are L_i-smooth, then f is L-smooth with $L \leq \frac{1}{n}\sum_{i=1}^n L_i$.

(ii) If f is L-smooth and f_i are convex, then f_i is L_i-smooth with $L_i \leq nL$.

(iii) If f is L'-smooth and f_i are convex and L_i'-smooth, then there exist $L \leq L'$ and $L_i \leq L_i'$ $\forall i$ such that f is L-smooth, f_i is L_i-smooth $\forall i$, and $L \leq \max_i L_i \leq nL$.

Proof of (iii)

Let's take

$$L = \min\Big(L',\ \inf\big\{L^* \in \mathbb{R}^+ : f \text{ is } L^*\text{-smooth}\big\}\Big)$$

and

$$L_i = \min\Big(L_i',\ \inf\big\{L_i^* \in \mathbb{R}^+ : f_i \text{ is } L_i^*\text{-smooth}\big\}\Big).$$

In that case, f is L-smooth, f_i is L_i-smooth, and $L_i \leq L_i'$, $L \leq L'$.

Now, using (i), we get that f is L''-smooth and $L'' \leq \frac{1}{n}\sum_i L_i$, but because $L \leq L''$, we get that

$$L \leq \frac{1}{n}\sum_{i=1}^n L_i \leq \max_i L_i.$$

Now, using (ii), we get that f_i is L_i''-smooth and $L_i'' \leq nL$, but because $L_i \leq L_i''$, we get that $L_i \leq nL$, so $\max_i L_i \leq nL$, which proves Lemma 44 (iii):

$$L \leq \frac{1}{n}\sum_{i=1}^n L_i \leq \max_i L_i \qquad \text{and} \qquad \max_i L_i \leq nL.$$

Exercise 35

Give an example of L_i-smooth functions f_i for which $\frac{1}{n}\sum_i L_i \ll \max_i L_i$.

Let $n = 1000$ and

$$f_1 = 500x^2, \qquad f_2 = x^2, \qquad f_i = \tfrac{1}{2}x^2 \quad \forall i = 3, \ldots, n,$$

so that $L_1 = 1000$, $L_2 = 2$ and $L_i = 1$ for $i \geq 3$. In this case

$$\frac{1}{n}\sum_{i=1}^n L_i = \frac{1}{1000}\cdot\Big(1000 + 2 + \sum_{i=3}^{1000} 1\Big) = \frac{2000}{1000} = 2$$

and $\max_i L_i = 1000$, so we have

$$\max_i L_i = 1000 \gg 2 = \frac{1}{n}\sum_{i=1}^n L_i.$$

Exercise 36

Let $a_1, \ldots, a_n > 0$. Prove that the solution to the optimization problem

$$D_{\Delta_n} := \min_{p \in \Delta_n} \max_i \frac{a_i}{p_i}$$

is given by $p_i = \frac{a_i}{\sum_j a_j}$ for all $i = 1, \ldots, n$.

Proof

First of all, note that for $p_i = \frac{a_i}{\sum_j a_j}$ we have

$$D := \max_i \frac{a_i}{p_i} = \max_i\Big(\sum_j a_j\Big) = \sum_j a_j,$$

i.e. all the ratios $\frac{a_i}{p_i}$ are equal to the same number $\sum_j a_j$.

Let's take another $p' \in \Delta_n$ and calculate $D' = \max_i \frac{a_i}{p'_i}$. If $p' \neq p$, then there is an i with $p'_i \neq p_i$. If $p'_i < p_i$, then let's define $k = i$, and if $p'_i > p_i$, then, since $\sum_j p'_j = \sum_j p_j = 1$, there is $j \neq i$ with $p'_j < p_j$, and we will define $k = j$. So in both cases there is a k with $p'_k < p_k$, and in that case

$$D = \sum_j a_j = \frac{a_k}{p_k} < \frac{a_k}{p'_k} \leq \max_i \frac{a_i}{p'_i} = D'.$$

So for any $p' \neq p$ we have $D < D'$, which means that p is the (unique) solution.

Exercise 37 (Importance sampling 2 – simple version)

Let $a, b > 0$. Prove that the solution to the optimization problem

$$\min_{0<p<1} \frac{a}{p} + \frac{b}{1-p}$$

is given by $p = \frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}$.

Proof

Let $\varphi(p) := \frac{a}{p} + \frac{b}{1-p}$ for $p \in (0,1)$. Then

$$\varphi'(p) = -\frac{a}{p^2} + \frac{b}{(1-p)^2}, \qquad \varphi''(p) = \frac{2a}{p^3} + \frac{2b}{(1-p)^3} > 0,$$

so φ is strictly convex on $(0,1)$, and a point where φ' equals 0 is its unique minimiser. Equating the derivative to 0 and using $p > 0$, $1 - p > 0$ to take square roots, we get

$$\varphi'(p) = 0 \iff \frac{a}{p^2} = \frac{b}{(1-p)^2} \iff \frac{\sqrt{a}}{p} = \frac{\sqrt{b}}{1-p} \iff \sqrt{a}\,(1-p) = \sqrt{b}\,p \iff p = \frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}.$$

Since $a, b > 0$, this p lies in $(0,1)$, so it is the solution. The optimal value is

$$\varphi\left(\frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}\right) = \sqrt{a}\left(\sqrt{a}+\sqrt{b}\right) + \sqrt{b}\left(\sqrt{a}+\sqrt{b}\right) = \left(\sqrt{a}+\sqrt{b}\right)^2.$$